

Experimental Study on the Performance Enhancement of Solar Air Heaters Using Passive Flow Controllers with Louvered Sheets

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Why improve solar air heaters?

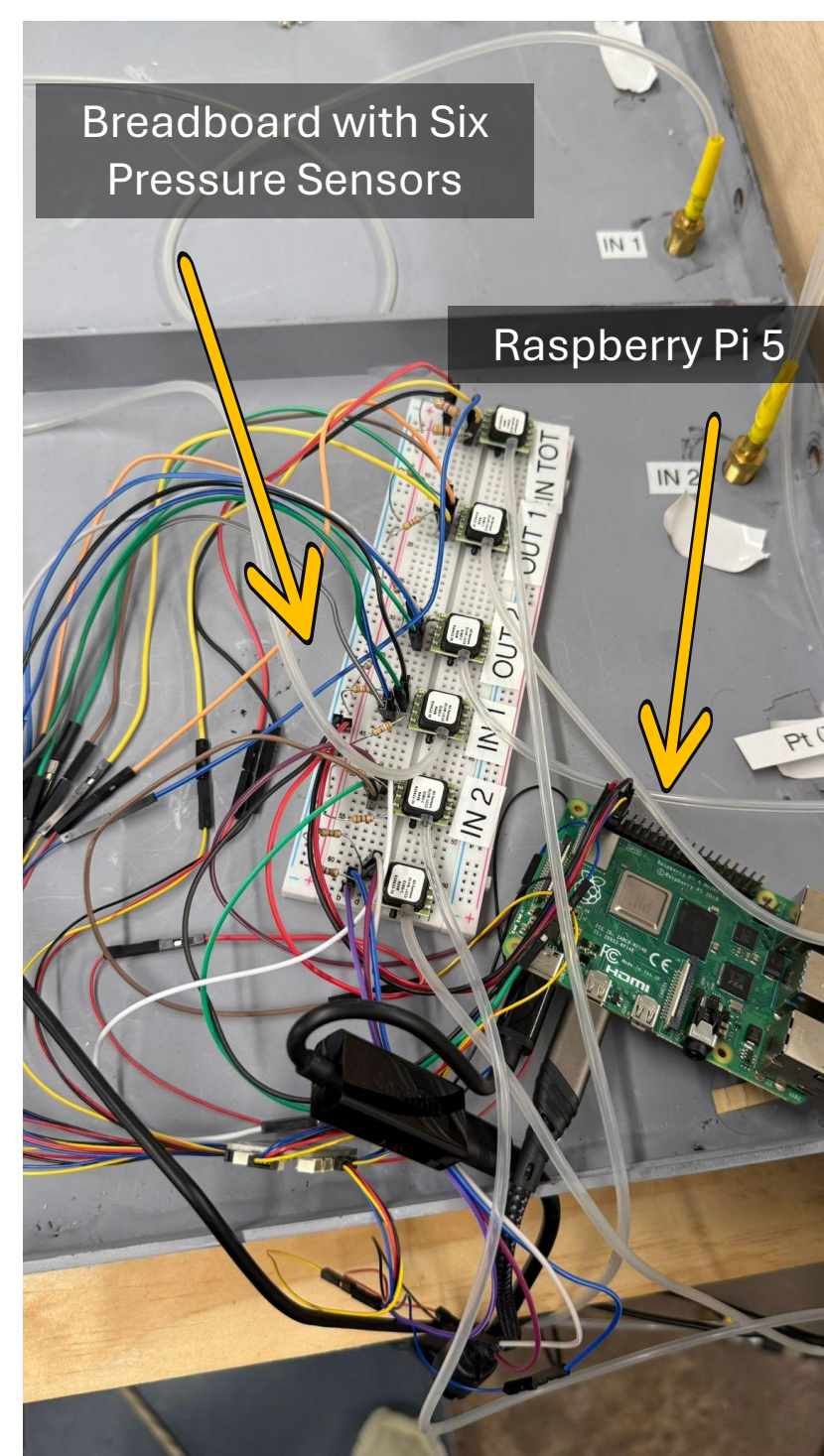
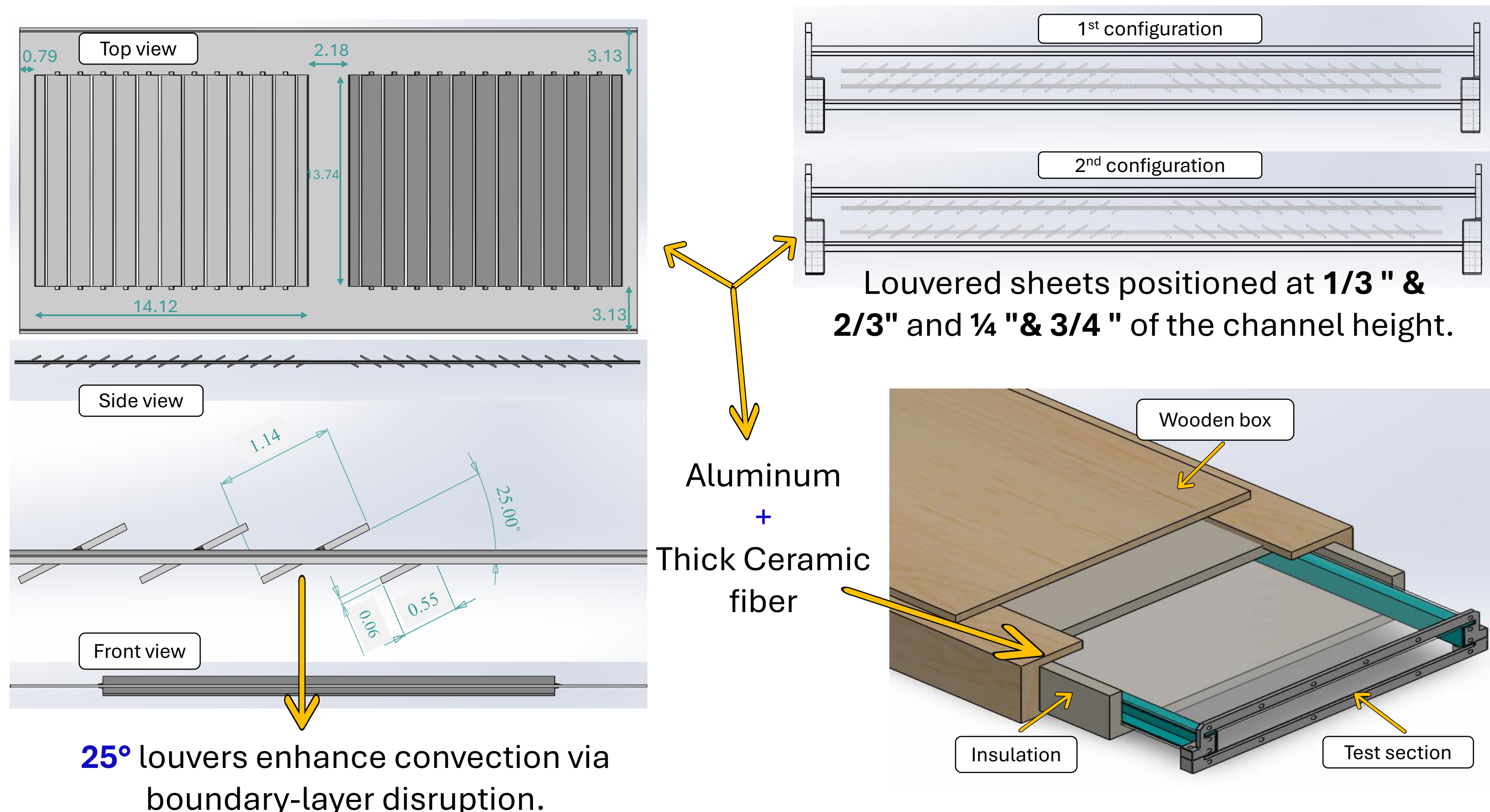
- Solar air heaters waste a significant portion of available solar energy due to **weak convective** heat transfer.

- Improving convection increases usable energy output without increasing solar input.

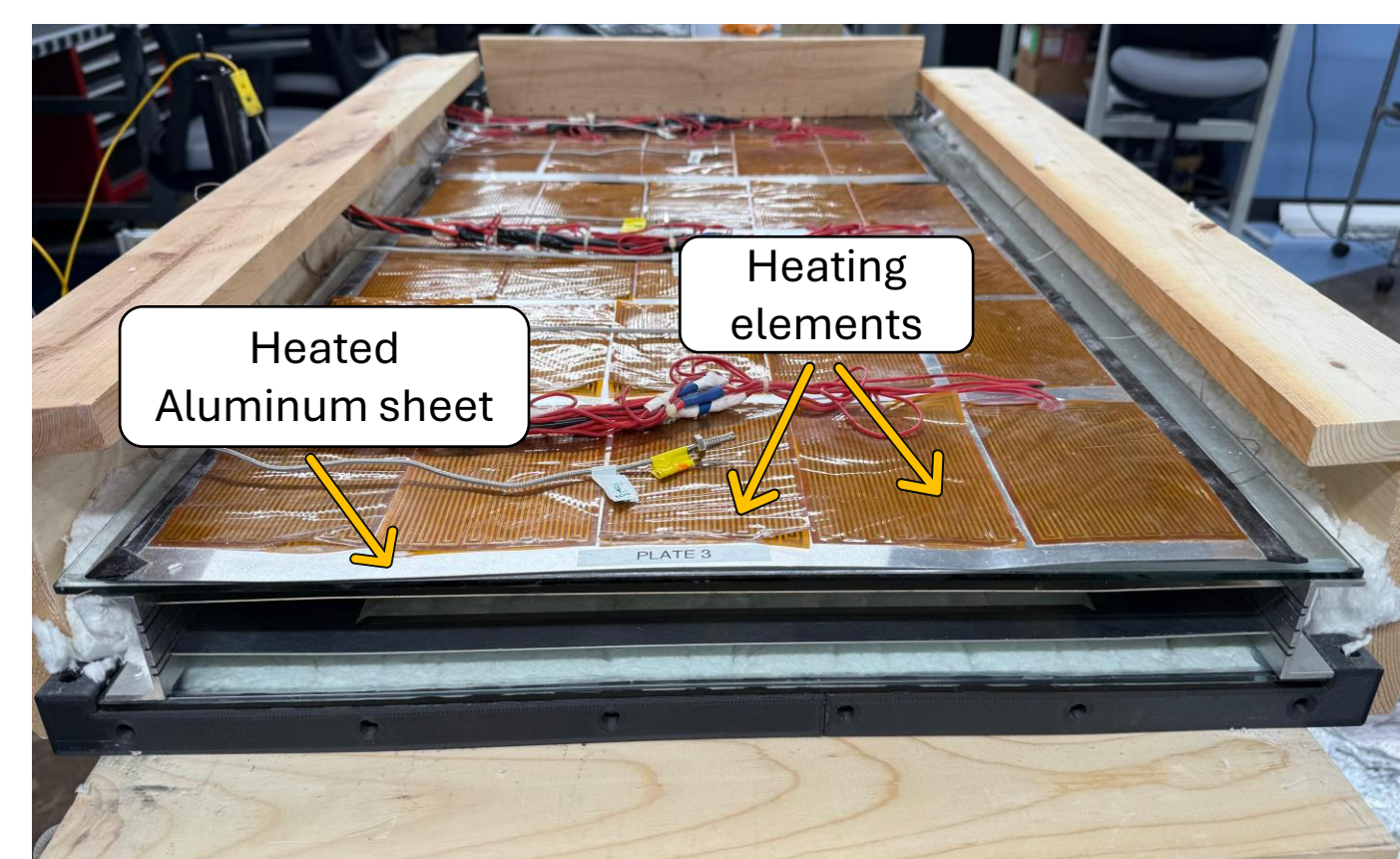
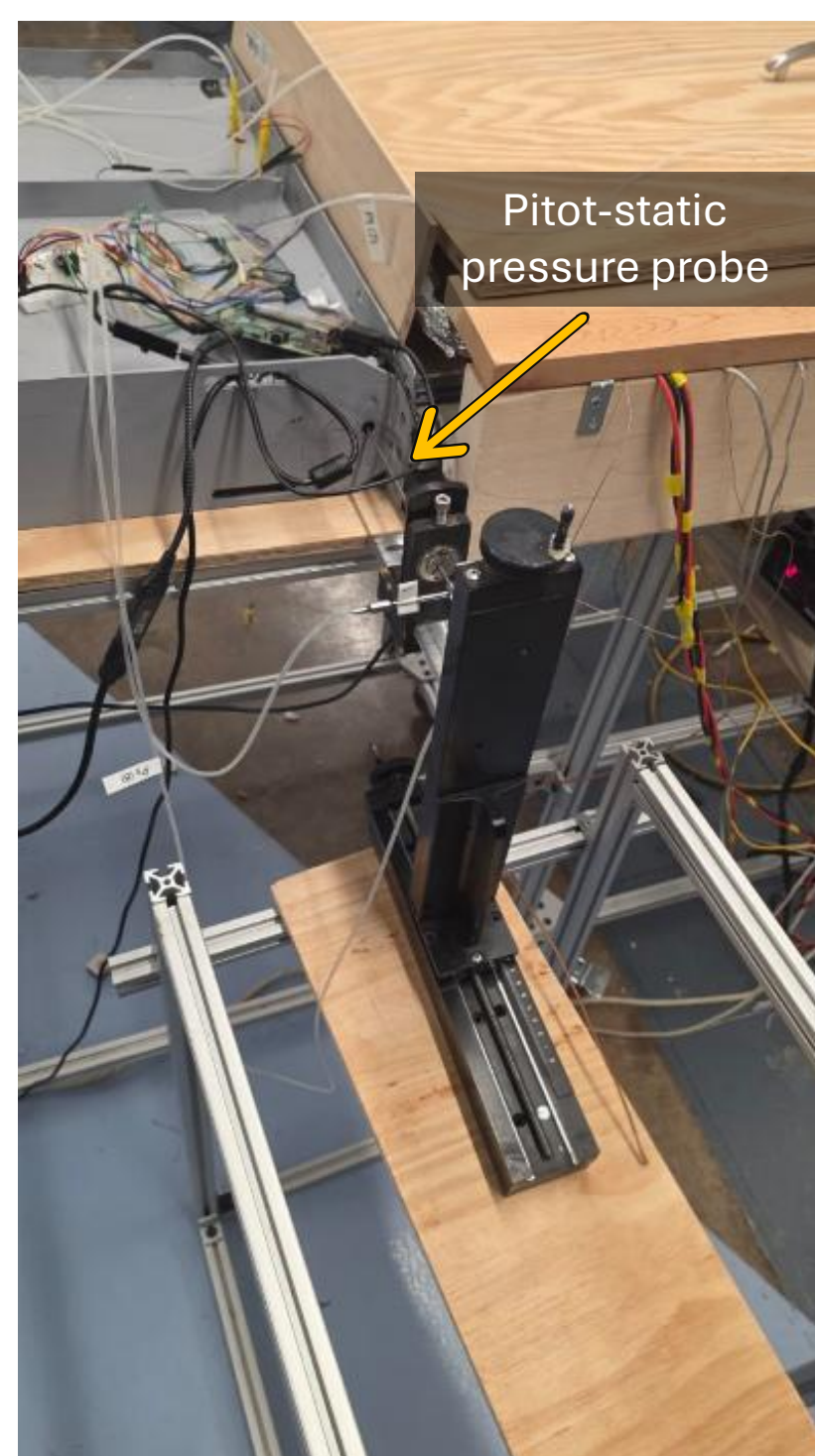
+74% Thermo-Hydraulic Performance possible

Low heat transfer coefficient + Thermal boundary layer → Reduced Convection (Low efficiency)

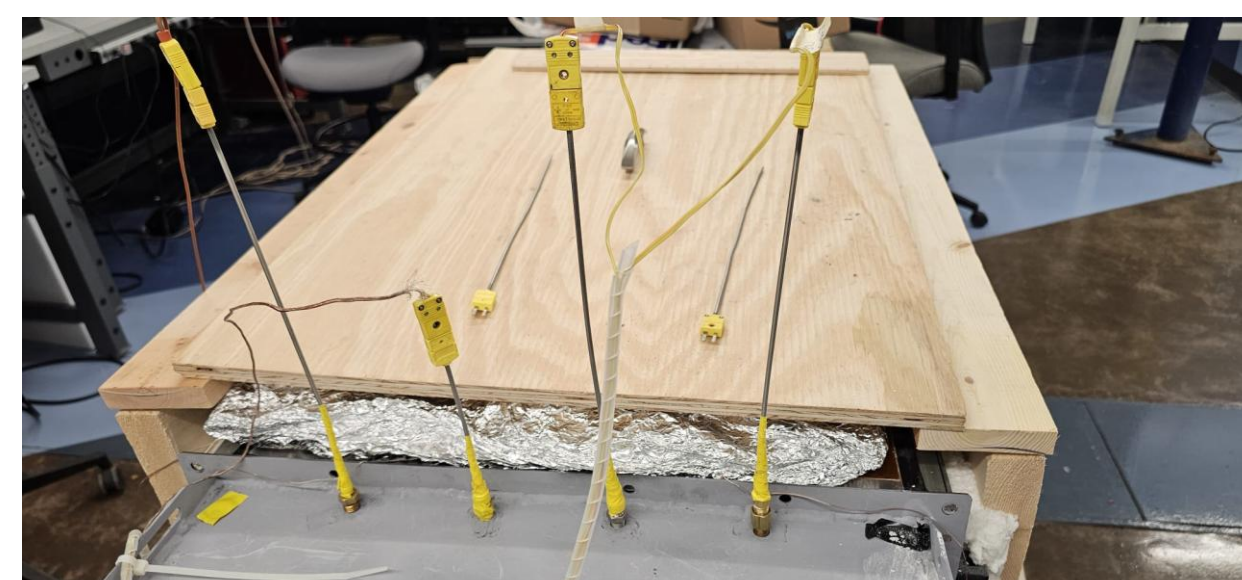
Test Configuration & Data Collection



Differential pressure measured using an **Amphenol DLHR-L01D** sensor with **Raspberry Pi**-based data acquisition.



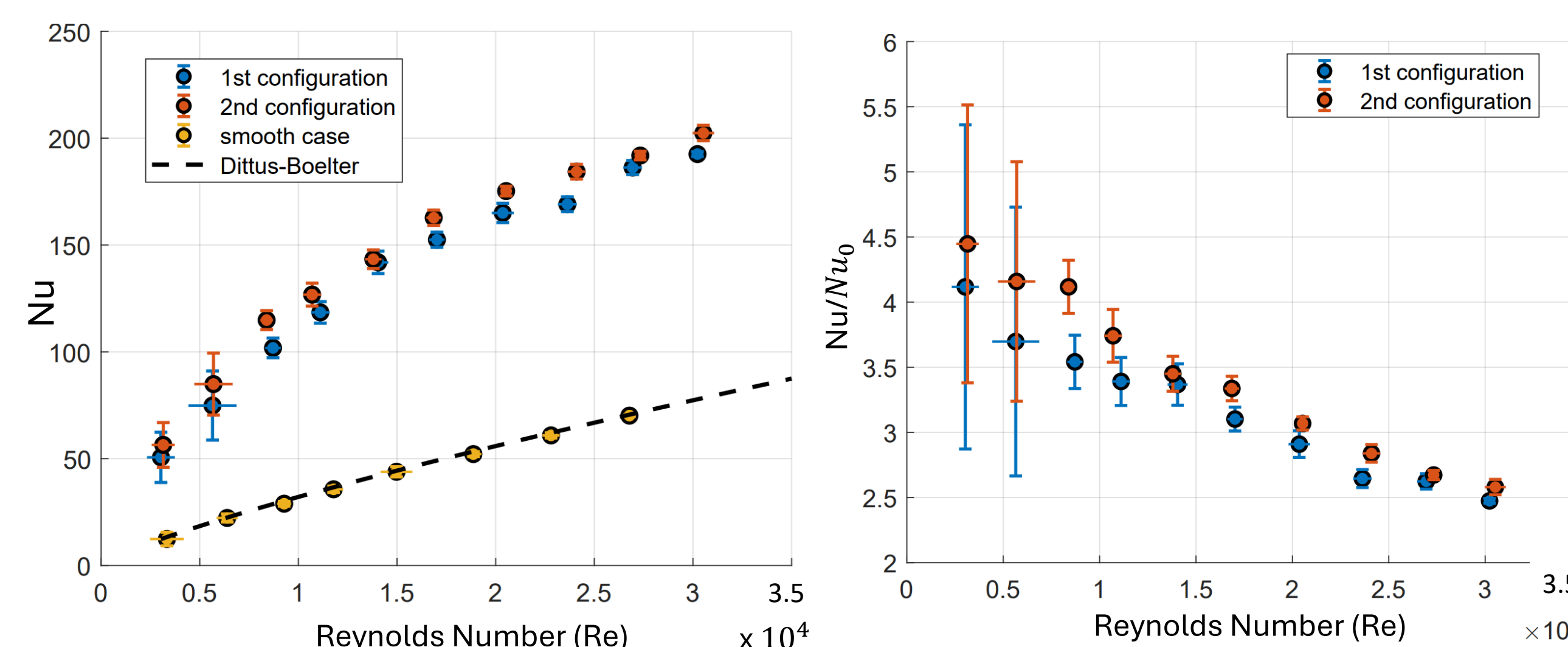
Insulated solar air heater duct with uniform electric heating ($6 \times 50 \text{ W}$) under steady-state airflow.



Experimental Validation & Performance Enhancement

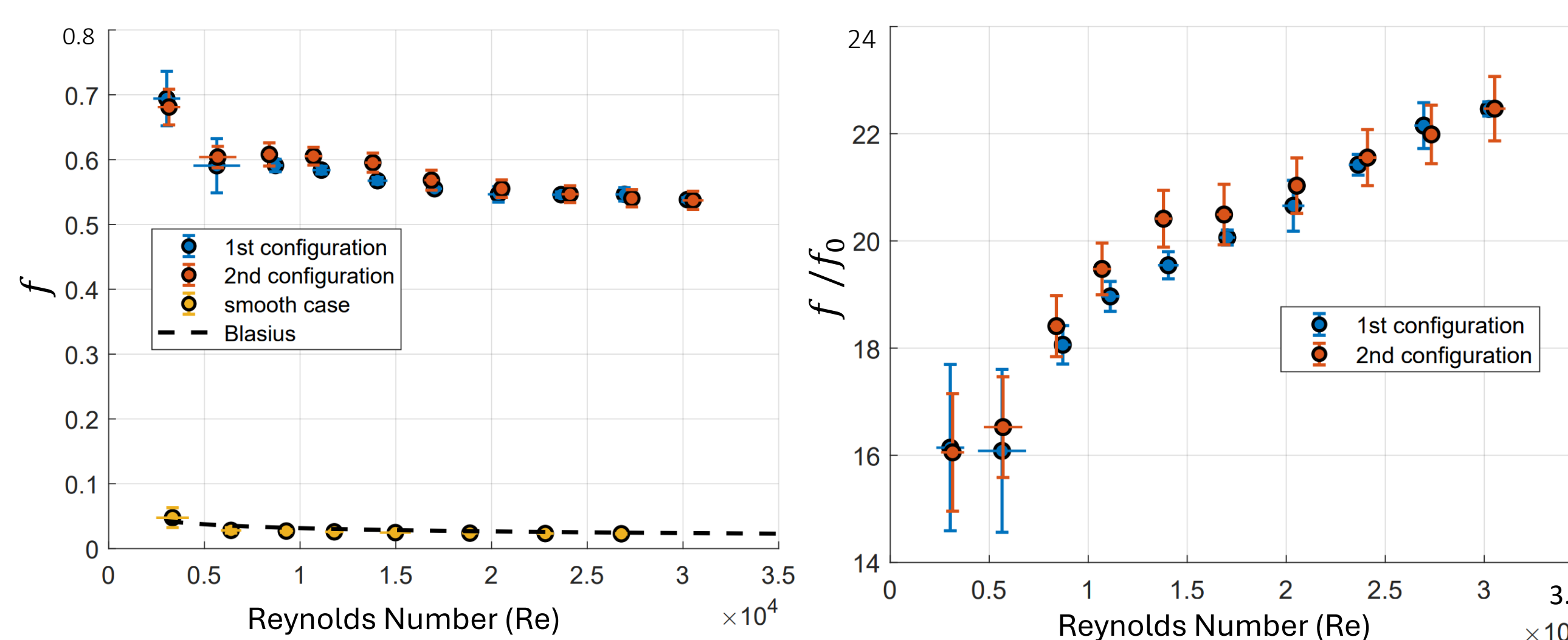
Heat Transfer Enhancement

Around 4x increase in **Nusselt number (Nu)** at lower flow rates compared to the smooth channel



Hydraulic Performance

Configuration 2 (sheets at $\frac{1}{4}$ & $\frac{3}{4}$ height) is superior, reaching a peak **THPF of 1.74**



Important Equations

$$\dot{m} = \rho A_c U \quad Nu_{calc} = 0.023 Re^{0.8} Pr^{0.4}$$

$$Q_{conv} = Q_{air} = \dot{m} C_p (\bar{T}_{out} - \bar{T}_{in})$$

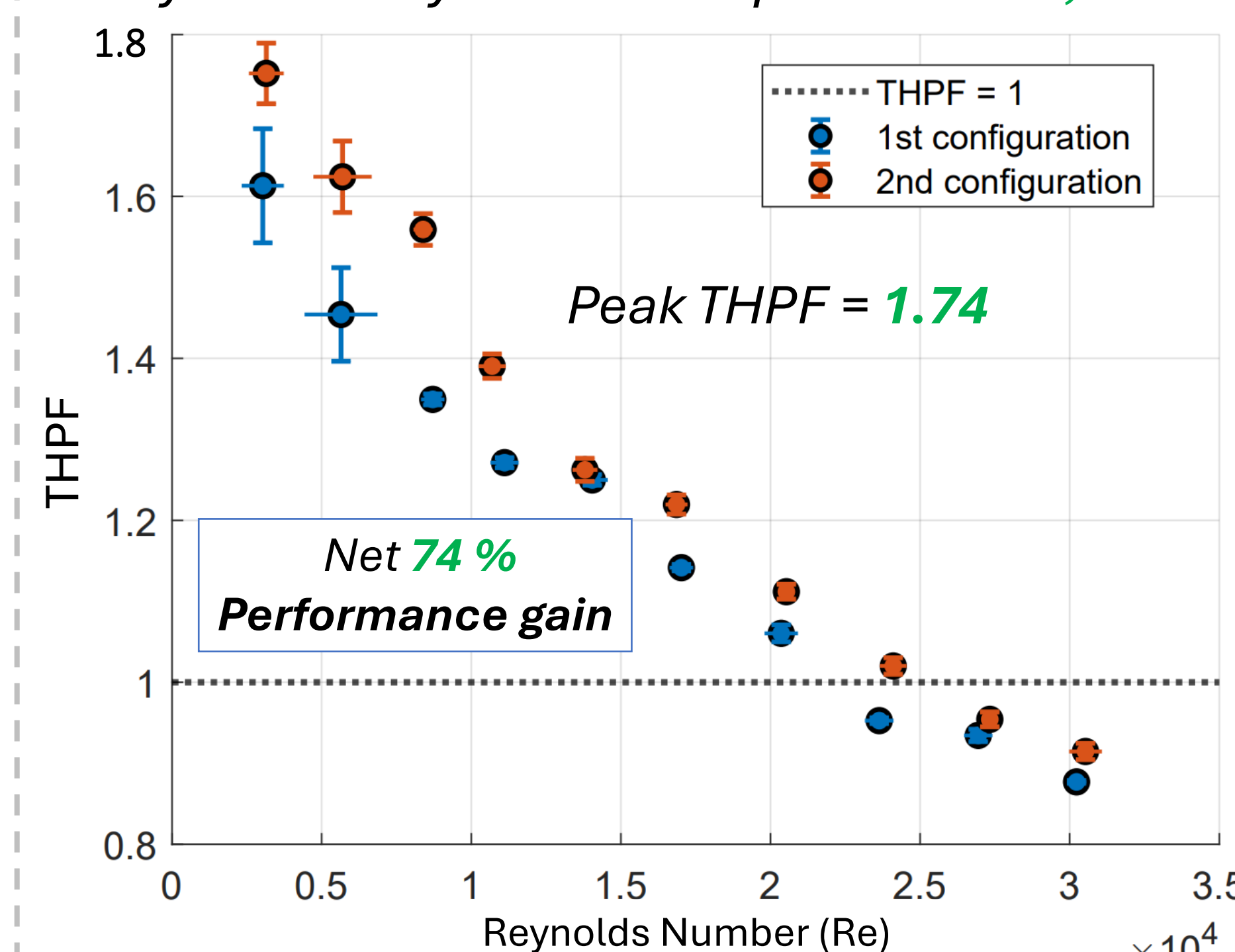
$$\bar{T}_f = \frac{\bar{T}_{out} + \bar{T}_{in}}{2} \quad F_R = \frac{f}{f_0} \quad Nu = \frac{h D_h}{k}$$

$$D_h = \frac{4 A_t}{p_{wet}} \quad Nu_R = \frac{Nu}{Nu_0}$$

$$h = \frac{Q_{conv}}{A(\bar{T}_s - \bar{T}_f)} \quad THPF = \frac{Nu_R}{f_R^{1/3}}$$

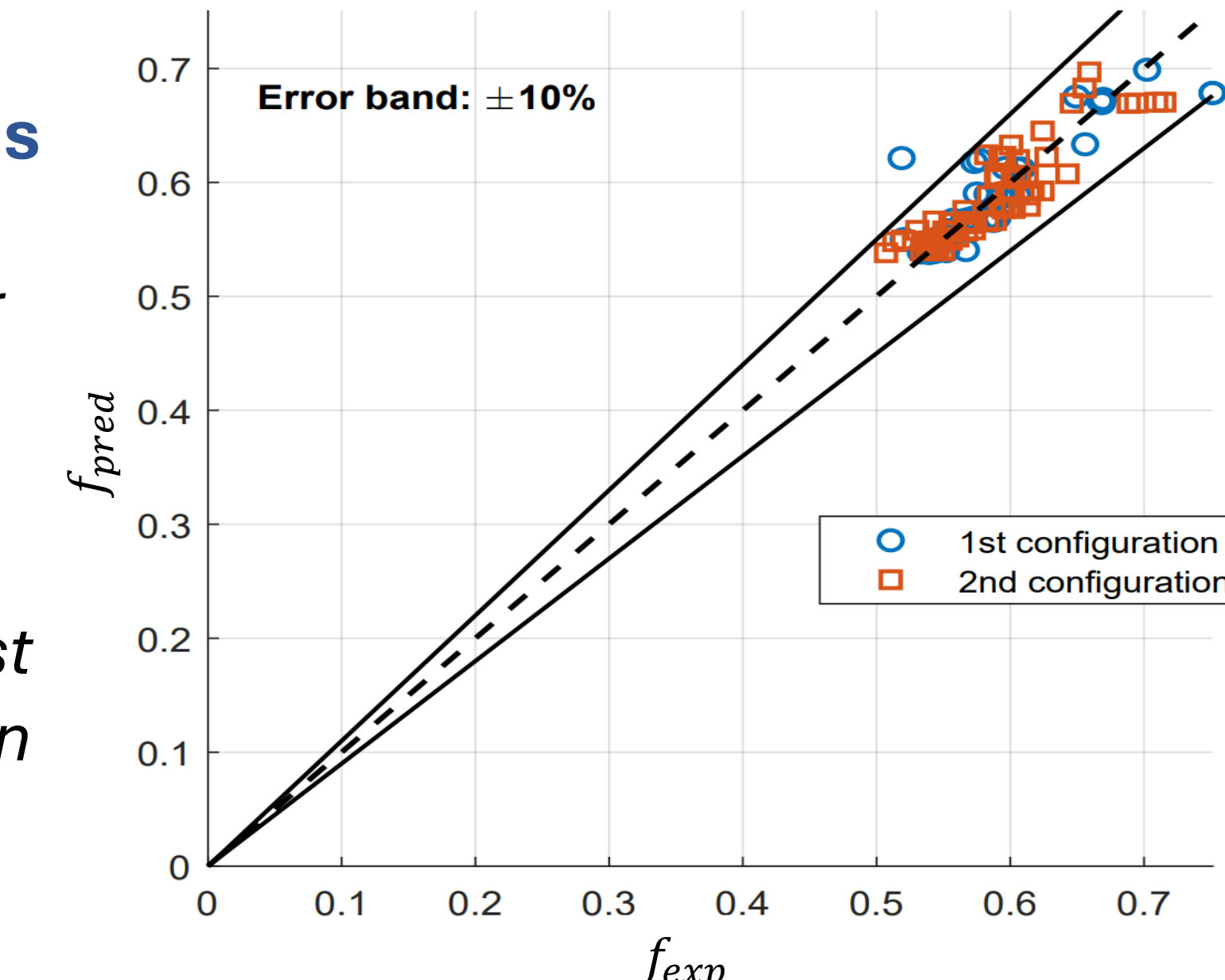
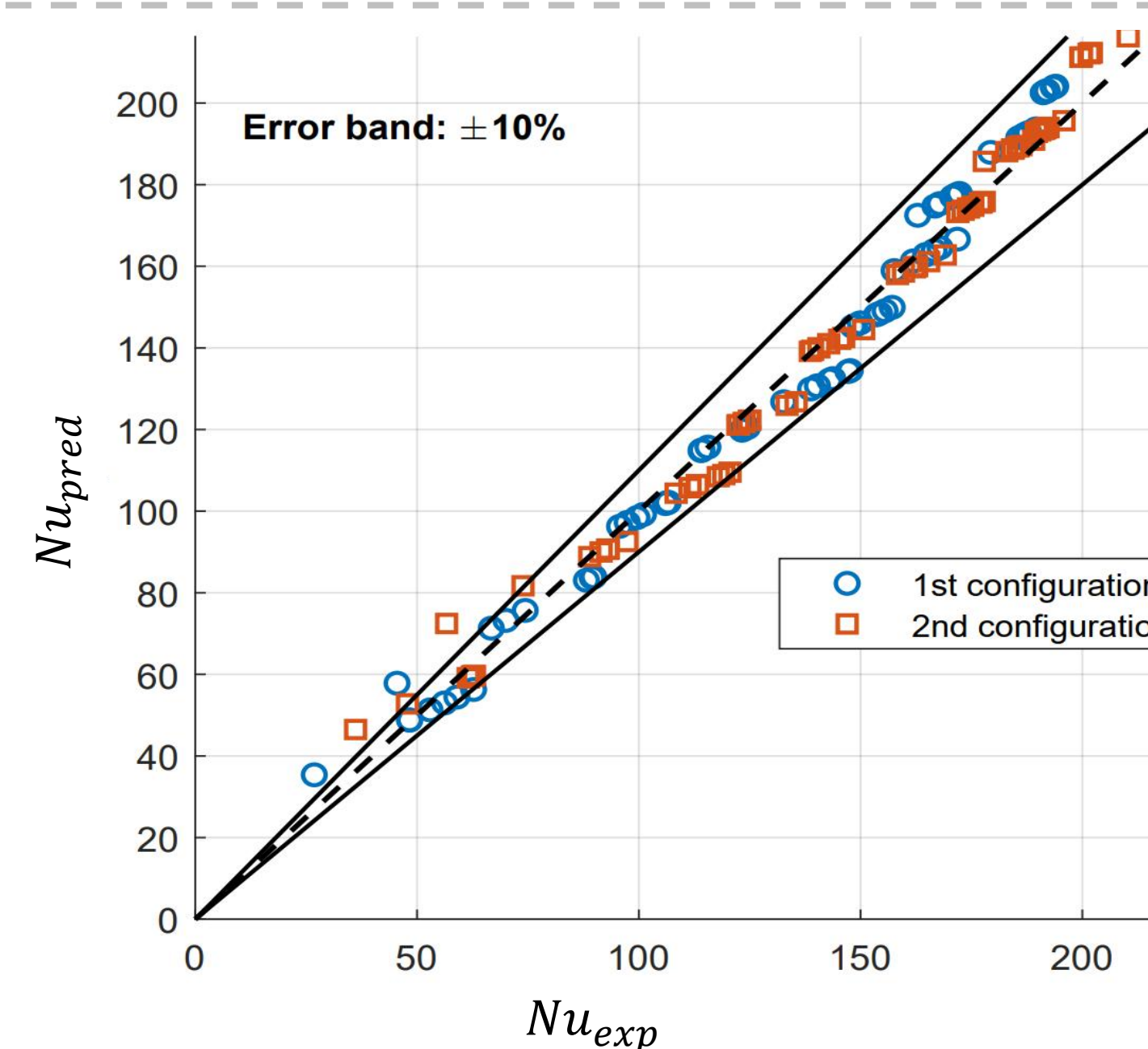
Thermo-Hydraulic Performance Factor (THPF)

Performance remained thermally and hydraulically favorable up to $Re \approx 24,000$



Model Validation & Error Analysis

- Data validated within $\pm 10\%$ error bands against MATLAB empirical correlations.
- Demonstrates high setup **reliability and accuracy**, with most data points falling within deviation bands.



Conclusion & Future Work

- This experimental setup supports the development of optimized solar air heater designs that improve heat transfer while minimizing losses
- Deeper understanding of boundary-layer dynamics will guide the influence of key design parameters
- Correlations for Nusselt number and friction factor are being developed
- Particle Image Velocimetry (PIV) measurements are planned to further investigate boundary-layer behavior

References

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- Blasius, H. (1913). Das Ähnlichkeitsgesetz bei Reibungsvorgängen in Flüssigkeiten, Forschungsarbeiten auf dem Gebiete des Ingenieurwesens, 131
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